

Circuit Simulation Lab:14

T Flip-flops

Introduction

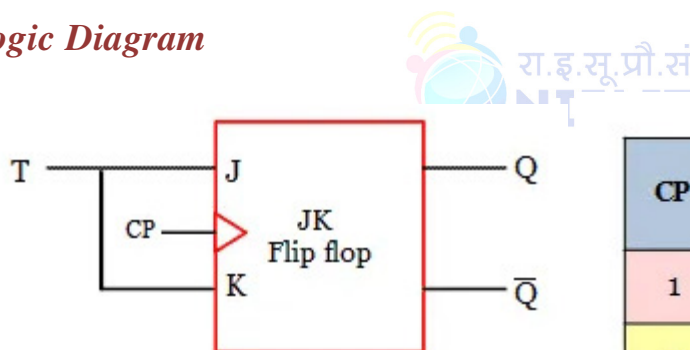
The purpose of this experiment is to introduce the design of simple sequential circuits, in this case T Flipflop. T flip flop is a modification of JK flip-flop. The J and K inputs are connected together to get the T input of flip flop. It is also called as Toggle flip flop.

Software tools Requirement

Modelsim (Siemens)

Xilinx Vivado

Logic Diagram



CP	T	Q_{n+1}	State
1	0	Q_n	NO CHANGE
1	1	$\bar{Q}_n$	TOGGLES

J-K Flip Flop

Truth Table

Describe the above circuits in Verilog HDL and capture the Waveforms

1. *Behavior modeling*
2. *Structural modeling (Use the JK Flip Flop for designing the T FF)*